Proper random walk model of order 2 (PRW2)

Parametrization

This is the proper version of the RW2 prior, defined for regular locations $1, 2, \ldots$,

$$x_t = 2\gamma x_{t-1} - \gamma^2 x_{t-2} + \epsilon_t, \qquad |\gamma| < 1$$

where ϵ_t is iid zero mean Gaussian noise. This process is made stationary with marginal precision κ, and range r, which is the 'distance' or 'time' to small correlation, is defined (implicitely) as

$$\rho = \exp\left(-\frac{\sqrt{12}h}{r}\right)$$

where h is (typical) distance or time, 'step-size', between the locations where the process is defined (here $h = 1$). The correlation function of the process is found to be

$$\rho(h) = \left(1 + h\frac{1-\gamma^2}{1+\gamma^2}\right)\gamma^h, \qquad h = 0, 1, 2, \ldots$$

Similar to RW1 and RW2, this model allows for irregular locations (`loc`) and the process is then derived from the corresponding limiting process in continous time.

Note that r and h is in real scale, ie the scale of `loc`.

Hyperparameters

This model has two hyperparameters, the marginal precision κ and the range r. These are defined as

$$\theta_1 = \log(\kappa)$$
$$\theta_2 = \log(r)$$

and the priors are given for (θ_1, θ_2).

Specification

This model is given as `f(loc, model="prw2", ...)`.

Note that `constr=FALSE` by default for this model, unlike the similar but intrinsic, RW2 model.

The default is use PC-priors for both parameters, see `inla.doc("pc.prec")` and `inla.doc("pc.prw2.range")`. The prior `pc.prw2.range` is currently fixed and not possible to replace, as this prior is viral for the PRW2 model construction and motivation itself.

Hyperparameter spesification and default values

doc `Proper random walk of order 2`

hyper

theta1

hyperid `6103`
name `log precision`
short.name `prec`
prior `pc.prec`
param `1 0.01`
initial `4`

fixed FALSE

to.theta function(x) log(x)

from.theta function(x) exp(x)

theta2

hyperid 6102

name log range

short.name range

prior pc.prw2.range

param 0 0 0 0

initial 3

fixed FALSE

to.theta function(x) log(x)

from.theta function(x) exp(x)

constr FALSE

nrow.ncol FALSE

augmented FALSE

aug.factor 1

aug.constr

n.div.by

n.required FALSE

set.default.values FALSE

pdf prw2

Example

```
n <- 200
loc <- seq(0, n-1, length.out=n)
h.size <- diff(range(loc)) / (n-1)
f.true <- (sin(2*pi*(loc/n)^2))^3
s = 0.2
y <- f.true + rnorm(n, sd = s)

plot(loc, y, pch = 19)
lines(loc, f.true, type='l', lwd = 3, col='blue')

r <- inla(y ~ -1 + f(loc, model = "prw2", values = loc,
                     hyper = list(
                         range = list(
                             param = c(50, 0.5, h.size, 0)))),
          family = "normal",
          control.family = list(
              hyper = list(prec = list(initial = log(1/s^2),
                                       fixed = TRUE))),
          data = data.frame(y, loc))
lines(loc, r$summary.linear.predictor$mean, lwd = 3, col = "red")
```

```
## we can now change the resolution and get the 'same' results
nn <- 2*n
lloc <- seq(0, n-1, length.out = nn)
yy <- numeric(nn)
for(i in 1:n) {
    yy[1 + (i-1) * 2] <- y[i]
    yy[2 + (i-1) * 2] <- NA
}
hh.size <- diff(range(lloc)) / (nn-1)

rr <- inla(yy ~ -1 + f(lloc, model = "prw2", values = lloc,
                       hyper = list(
                           ## note that '50' is the same as its in the real scale
                           range = list(param = c(50, 0.5, hh.size, 0)))),
           family = "normal",
           control.family = list(hyper = list(prec = list(initial = log(1/s^2),
                                                          fixed = TRUE))),
           data = data.frame(yy, lloc))
lines(lloc, rr$summary.linear.predictor$mean, lwd = 3, col = "red")

## once more
nnn <- 4*n
llloc <- seq(0, n-1, length.out = nnn)
yyy <- numeric(nnn)
for(i in 1:n) {
    yyy[1 + (i-1) * 4] <- y[i]
    yyy[2:4 + (i-1) * 4] <- NA
}
hhh.size <- diff(range(llloc)) / (nnn-1)

rrr <- inla(yyy ~ -1 + f(llloc, model = "prw2", values = llloc,
                       hyper = list(
                           ## again, '50' is not changed
                           range = list(param = c(50, 0.5, hhh.size, 0)))),
           family = "normal",
           control.family = list(hyper = list(prec = list(initial = log(1/s^2),
                                                          fixed = TRUE))),
           data = data.frame(yy, lloc))
lines(llloc, rrr$summary.linear.predictor$mean, lwd = 3, col = "red")

inla.dev.new()
res <- cbind(r$summary.linear.predictor$mean,
             rr$summary.linear.predictor$mean[seq(1, nn, by = 2)],
             rrr$summary.linear.predictor$mean[seq(1, nnn, by = 4)])
pairs(res)

inla.dev.new()
plot(r$marginals.hyperpar[[2]],  pch = 19)
lines(rr$marginals.hyperpar[[2]], lwd = 3, col = "blue")
lines(rrr$marginals.hyperpar[[2]], lwd = 3, col = "red")
```

Notes

The details for this model is available in the PhD thesis with link `http://hdl.handle.net/10754/707346`